

Modeling Human Activity States Using Hidden Markov Models

Formative 2 | Machine Learning Operations | African Leadership University

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1. Background and Motivation

Wearable health monitors, fitness trackers, and smart home systems must recognize physical activity in real time from raw sensor data without any user input. A Hidden Markov Model (HMM) works well for this problem. The actual activity, whether someone is standing, walking, or jumping, is never directly seen. The only information available is the continuous, noisy output from accelerometer and gyroscope sensors. An HMM addresses this by separating the hidden activity state from the observable sensor signal. It captures two important aspects of human movement: the pattern each activity creates in the sensor data (emission probabilities) and how activities realistically shift from one to another over time (transition probabilities). This project creates a complete HMM-based activity recognition pipeline, covering data collection, Viterbi decoding, and evaluation on withheld recordings, using four activities: Still, Standing, Walking, and Jumping.

2. Data Collection and Preprocessing

Data was collected using an iPhone with the Sensor Logger app by Chio Tsz Hei. The Accelerometer (x, y, z in g) and Gyroscope (x, y, z in rad/s) both operated at 50 Hz. The phone was held at waist level for Standing, Walking, and Jumping, and placed flat on a surface for Still. Each activity was recorded fifteen times, with each recording lasting about ten seconds. This totaled about 150 seconds per activity and ten minutes overall. Files were saved with timestamps and organized into folders by activity. Since only one phone and one person were involved, there was no need to match sampling rates across devices. Four extra recordings, one for each activity, were made in a separate session and kept entirely away from training. The Accelerometer and Gyroscope files were then combined by timestamp into one six-column table for each recording. The full training set contains 30,821 rows across 60 recordings: Jumping 7,766, Standing 7,726, Walking 7,689, Still 7,640. Figure 1 displays the raw Accelerometer signal for one recording per activity, which influenced the feature choices in Section 3.

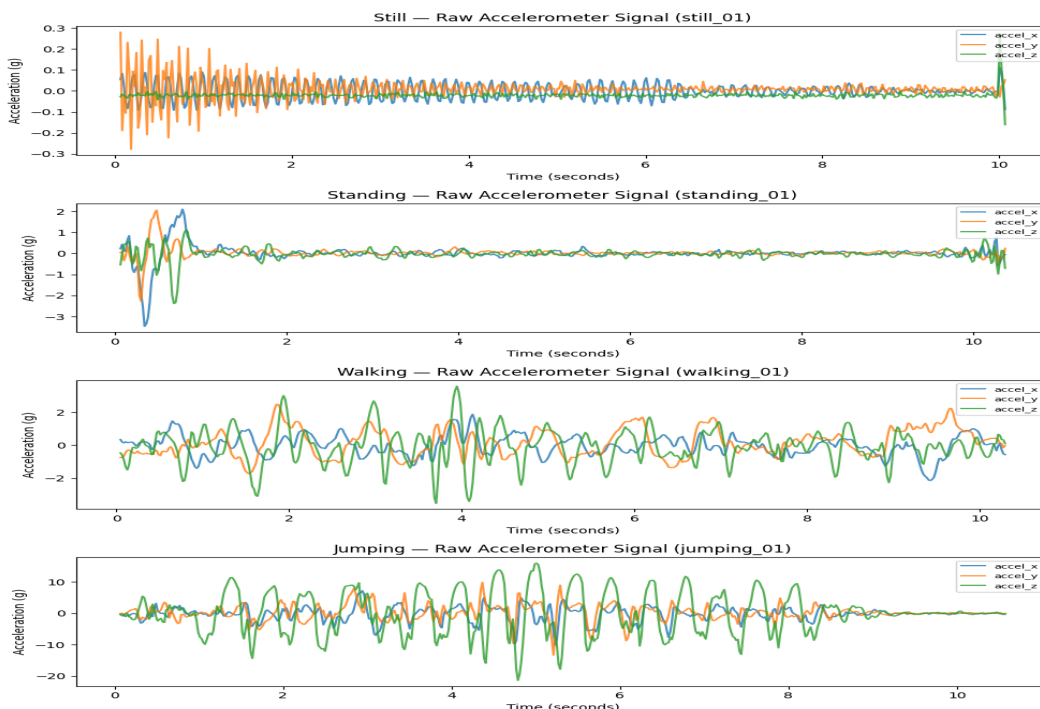


Figure 1. Raw accelerometer signals (*accel_x*, *accel_y*, *accel_z*) for one 10-second take per activity at 50 Hz.

3. HMM Setup and Implementation

3.1 Feature Extraction and Windowing

Each take was divided into non-overlapping 1-second windows of 50 samples, at a rate of 50 Hz. This length is enough to capture a complete walking stride or jump arc but short enough to avoid including activity transitions. This resulted in 599 training windows, with about 150 for each activity. Sixteen features were extracted from each window

- Time-domain (14): mean and standard deviation per axis for both the accelerometer and gyroscope (12 features); Signal Magnitude Area (SMA) for the accelerometer and gyroscope (2 features). The mean shows directional bias, the standard deviation measures variability, and SMA gives a scalar energy measure that increases steadily from Still to Jumping.
- Frequency-domain (2): dominant frequency and spectral energy, which come from the FFT of the accelerometer magnitude signal after DC removal. The dominant frequency shows the stride cadence of Walking (about 1.5 to 2 Hz) and Jumping (about 2 to 4 Hz), but is absent in Still and Standing. Spectral energy measures total signal power, which is highest for Jumping due to its impulsive waveform.

All 16 features were normalized using Z-score, based on statistics from the training data. These statistics were then applied to the test set without any changes. Normalization is essential because feature scales vary widely (standard deviations are in single digits while spectral energy can reach thousands). This variation would otherwise skew HMM emission calculations toward high-magnitude features, regardless of their discriminative value.

3.2 Model Structure, Initialization, and Training

The Hidden Markov Model, or HMM, has four states that're not directly observable and uses 16 numbers to describe the emissions, which follow a simple bell-curve pattern. The HMM's transition matrix A was set up first.

The averages and spreads of emissions and the initial chances, denoted as π , were determined from labeled training data. We gathered counts of state transitions from sequences within takes. Adjusted them by adding 0.1 to ensure they are smooth. The emission parameters were calculated as the mean and spread for each class. A small value, epsilon of 0.000001, was used to maintain stability. This method of initialization helps avoid getting stuck in solutions, which often occurs with random initialization, especially when the dataset is small. The Viterbi algorithm works with logarithms to prevent small numbers. It uses a log-delta matrix and psi backpointers. The algorithm achieved an accuracy of 0.973 on the training data ranging from 0.900 to 1.000. The Baum-Welch EM refinement was run with a tolerance of 0.001 and a maximum of 50 iterations. During the E-step, gamma values were saved for reuse in the M-step variance update to ensure parameters. The training process converged in 7 iterations. The log-likelihood increased from 7,534 to 9,444, indicating convergence rather than just reaching the iteration limit. The accuracy after refinement is 0.962. In a validation test, hmmlearn's GaussianHMM with random initialization achieved a mapped accuracy of 0.733 under the same conditions. This shows that supervised initialization performs better than random initialization with this dataset. The HMM and its initialization method are effective for this task.

4. Results and Interpretation

4.1 Transition Matrix and Decoded Sequences

Figure 2 shows the learned transition matrix. The strong diagonal (0.923–1.000) reflects activity persistence within 1-second windows; Baum-Welch introduced plausible off-diagonal transitions (Standing→Walking: 0.015; Walking→Jumping: 0.028; Jumping→Walking: 0.066). Figures 3 and 4 show Viterbi-decoded training sequences: walking_01 is perfectly decoded across all 10 windows; still_08 shows a transient misclassification of Still as Standing in windows 0–2, attributable to residual vibration as the phone was placed onto the recording surface.

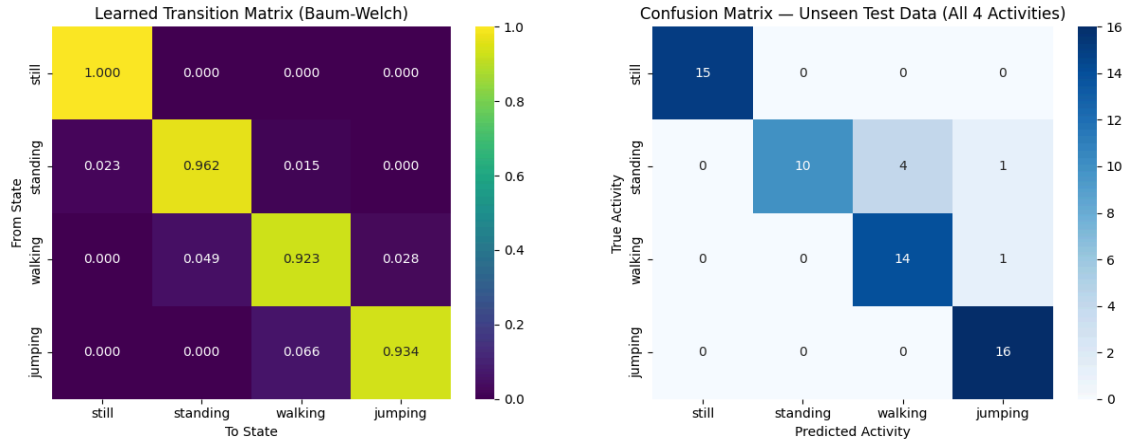


Figure 2 (left). Learned transition matrix. Figure 6 (right). Confusion matrix on unseen test data.

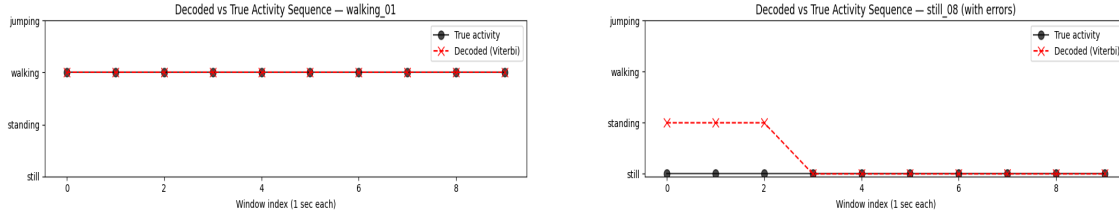


Figure 3 (left). Decoded vs. true sequence for walking_01 — all windows correct. Figure 4 (right). still_08 — transient error windows 0–2.

4.2 Emission Distributions

Figure 5 shows what we learned about the emission means for each hidden state. Jumping stands out because it has variance, SMA, and spectral energy values that are quite high around 1.4 to 1.6. Still and Standing have similar low-energy profiles, which is why they sometimes get confused. Walking is somewhere in between. The dominant frequency feature helps tell them apart from Walking even though their raw energy levels are similar. This shows that using features from the frequency domain is useful. The values are 1.46 for Still and -0.64 for Walking.

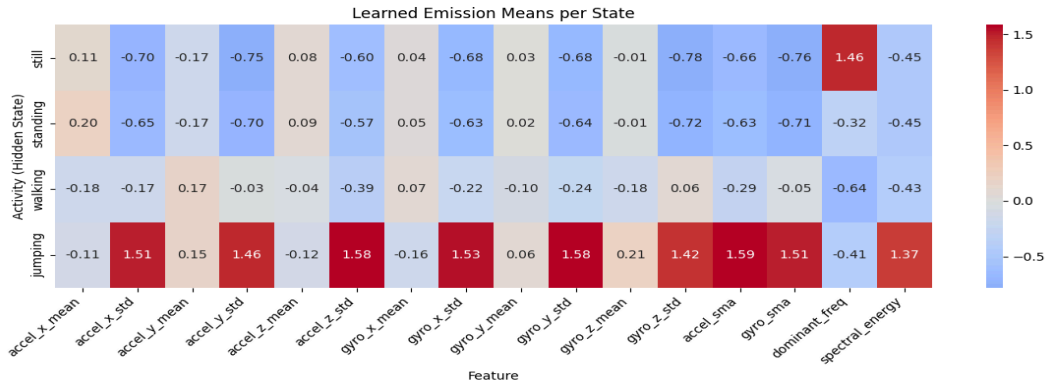


Figure 5. Learned emission means per hidden state (normalized Z-scores). Jumping is strongly differentiated; Still and Standing share overlapping low-energy profiles.

4.3 Evaluation on Unseen Data

The model was tested on four recordings it had never seen before, collected in a separate session after training was complete. It did a good job overall, getting things right 90.2% of the time. Looking at each activity individually, the model was perfect at recognizing Still and Jumping, getting them right every single time (sensitivity of 1.000 for both). For Walking, it got it right most of the time but made one mistake: it predicted Jumping when the true activity was Walking, giving a sensitivity of 0.933. The hardest activity for the model to distinguish was Standing, where it only got it right 67% of the time (sensitivity 0.667). Four windows were predicted as Walking and one as Jumping when the true activity was Standing, which makes sense because the sensor patterns during Standing are similar to those during Walking and Jumping. Table 1 has the full breakdown of how the model performed on each activity, and Figure 6 shows the confusion matrix with all the mistakes laid out clearly.

Table 1. Per-activity evaluation metrics on unseen test data (overall accuracy: 0.902).

Activity	N Samples	Sensitivity	Specificity	Accuracy
Still	15	1.000	1.000	1.000
Standing	15	0.667	1.000	0.918
Walking	15	0.933	0.913	0.918
Jumping	16	1.000	0.956	0.967

5. Discussion and Conclusion

The Hidden Markov Model achieves 96.2 percent accuracy on the training data and 90.2 percent on the test data that it has not seen before. This means that a small set of data collected from smartphones is enough to train a classifier that can recognize what people are doing. The Hidden Markov Model can easily tell the difference between Jumping and Still because they are at the ends of the feature space. Walking is also easy to recognize because it has a frequency and a medium amount of movement. The main problem is Standing because its features are similar to Still when someone is resting and Walking when someone is swaying slightly. You can see this overlap in Figure 5. Figure 6. The Hidden Markov Model can predict how people move around in a way even though it was only trained on recordings of one activity at a time. There is a mistake when the Hidden Markov Model is trying to recognize still because of the vibration that happens when the smartphone is being placed. You can see this mistake in the recording called still_08. To make the Hidden Markov Model better, we could use features from the gyroscope, better probability models, train it on sequences of multiple activities, and ignore the beginning of Still recordings. The results show that the Hidden Markov Model can recognize what people are doing in a way even with minimal data collection if it is carefully set up with labeled sensor data from smartphones.

Appendix: Data Collection Summary

Contributor	Device	Sampling Rate	Takes / Activity
Glory Paul	iPhone (iOS)	50 Hz	15